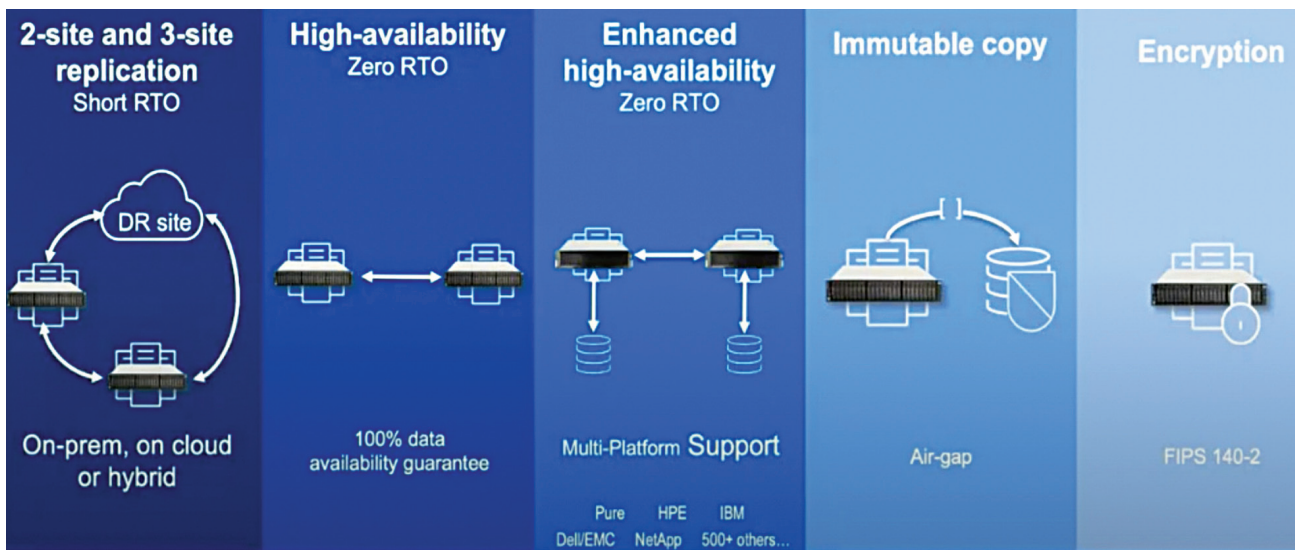




Characteristics of resilient primary storage

that he has been attacked, a lot of detective work is put in to find a clean uninfected copy. Traditionally, this has meant repeatedly mounting and inspecting progressively older backup copies.

“New systems make the copies searchable, or having e-discovery tools can aid you, so you mount the right recovery copy the first time. Choose tools that make the discovery process easy.”

5. Can data be restored fast? Discovering an attack takes a long time, which likely means the initial local snapshots would have migrated off to a secondary storage like an immutable object stored in the cloud, or a WORM tape. Hence, the most isolated copy might not be the fastest to restore.

“There is always a balance, and every organisation makes their own trade-off choices with a variety of storage media options that can be configured as isolated and immutable. You will have to consider the best option for your organisation.”

Characteristics of resilient primary storage

“Ensuring your data stays available to your applications is the primary function of storage.” So, what are the characteristics of resilient, primary storage that can help?

Remember, in a hybrid cloud reality, organisations are working with multiple locations, potentially mul-

tiple hardware vendors on premises and multiple public cloud providers. A resilient storage solution provides flexibility and helps in leveraging the infrastructure vendors and locations to create operational resiliency. This helps in achieving data resilience in the data centre and across virtualised, containerised, and hybrid cloud environments.

It ideally should be capable of traditional two-site and three-site replication configurations, including public cloud using a choice whether to have synchronous or asynchronous data communication. “This gives you confidence that your data can survive a localised disaster with very little or no data loss, also known as a recovery point objective or RPO.”

Resilient primary storage is also about gaining access to data quickly, in some cases immediately. This is known as a recovery time objective or RTO.

Resilient storage has options for immediate failover access to data at remote locations. “Not only does your data survive a localised disaster, but your applications have immediate access to alternate copies as if nothing ever happened.”

It is important to note that RPO and RTO options should be available regardless of the choice of primary storage hardware vendors, or public cloud providers—meaning multi-platform support. Resilient primary storage is also concerned with protec-

tion of data from bad actors or hackers. Encryption guards against prying eyes or outright data theft.

In the event of a primary copy of data becoming infected, resilient storage should also provide for making copies that are logically air-gapped from the primary data and further making that copy unchangeable or immutable.

Ensuring data resilience

Many organisations have a mix of different on-premise storage vendors or have acquired storage capacity over time, which means that they have different generations of storage systems. Throw in some cloud storage for a hybrid environment and you may find it quite difficult to deliver a consistent approach to data resilience. “The first step is modernising the storage infrastructure you already have.”

To conclude, the resiliency techniques can vary with the importance of the respective organisation’s workloads and resources to run the solution. It is wise for organisations with mission-critical workloads to utilise more resilient techniques at more levels, as the cost of not preserving the data would outweigh expenses during a prolonged outage. **EFY**

The article is based on a talk by Ron Riffe, Program Director, IBM Global Storage at SODACON 2021. It has been prepared by Abhinaya Kuzhanthaivel, Assistant Editor at EFY.